

Compute Before Retransmit: Reliability-Anchored Flow Matching for DeepJSCC HARQ

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Abstract—A quality-based hybrid automatic repeat request (HARQ) receiver requests another transmission when its current image estimate is inadequate. We investigate a preceding receiver action: repair selected latent tokens using the observation already available, and then decide whether physical retransmission is still needed. FlowHARQ combines reliability-anchored conditional flow matching with a separately calibrated quality predictor, without changing the source-channel codec. Under a two-round, temporally correlated Rayleigh model, the original three-training-run experiment and an existing ten-run extension show consistent aggregate savings on held-out DIV2K and Kodak24 images. In the extension, NACK probability falls by 1.319 and 1.453 percentage points, respectively; mean PSNR increases slightly, while LPIPS worsens within the specified tolerance. Paired decision analysis separates genuine saved rounds from quality changes on images that both receivers already accept. Charging 16-bit token scales and reliable ACK/NACK signaling reduces the corresponding payload savings of 0.758% and 0.846% to 0.353% and 0.391% at 0.5 information bits per complex channel use. The results establish a modest receiver-computation versus communication tradeoff, but substantial target-quality violations preclude a reliable-delivery claim or replacement for physical HARQ.

Index Terms—DeepJSCC, HARQ, flow matching, semantic communication, receiver processing.

I. INTRODUCTION

Deep joint source-channel coding (DeepJSCC) maps an image to channel symbols and reconstructs it without a separate bitstream decoder [1]. Its gradual quality degradation is useful on fading links, but does not remove the need to decide when reception is adequate. A quality-based HARQ receiver requests another transmission when a predicted reconstruction quality falls below a target. That decision consumes additional radio resources even when the receiver could still improve its estimate locally.

Existing retransmission mechanisms primarily change what is sent next. DeepJSCC-f exploits channel-output feedback to encode incremental information [2]; semantic-aware fine-grained HARQ (SAFG-HARQ) refines retransmission granularity [3]. In contrast, we ask whether receiver computation can avoid some requests *before* a second transmission is invoked. We call this action virtual retransmission, while emphasizing that it introduces no new channel observation and cannot replace missing physical evidence arbitrarily.

Generative refinement alone is not a new communication primitive. CDDM uses diffusion for channel denoising [4]. Recent flow-matching work includes FlowSem’s conditional image refinement [5] and channel-realization flow matching [6]; generative semantic HARQ has also been studied for text

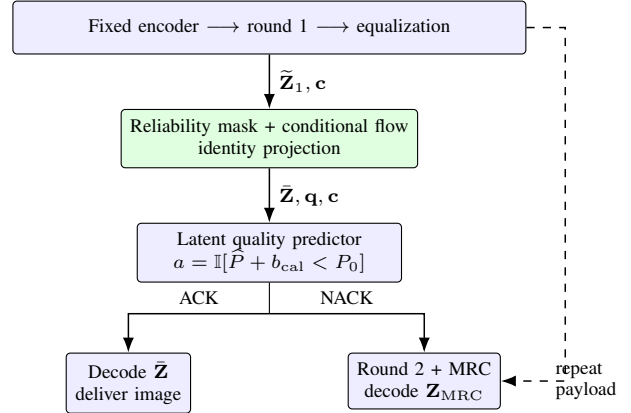


Fig. 1. Receiver computation precedes the physical retransmission decision. Both policies use the same first observation and identical MRC fallback. The dashed path carries the original payload, not newly generated information.

latents [7]. Our focus is the interaction between a constrained latent repair and the ensuing ACK/NACK decision, not the first application of flow matching to semantic communication. This distinction matters: a prettier reconstruction does not necessarily reduce transmissions, and a lower NACK rate could instead reflect an overconfident quality predictor.

FlowHARQ therefore couples three elements. A reliability head selects tokens for repair, an identity projection preserves the unselected latent values, and a separately calibrated quality head makes the final receiver-only decision. The encoder and decoder remain fixed. If the receiver still requests another round, it discards the repaired estimate and uses ordinary maximum-ratio combining (MRC) of two physical observations. This construction permits a paired comparison in which both policies have exactly the same fallback.

Our contributions are a reliability-anchored receiver formulation; a controlled rate-quality evaluation with explicit original and expanded experimental cohorts; and a decision-level and signaling-cost audit of the measured gain. The latter is important for reproducibility: the codec normalizes each token using an image-dependent scale, which is side information rather than free receiver knowledge. We charge that information explicitly and distinguish percentage-point NACK reductions from relative channel-use savings. The resulting improvement is small but interpretable. We do not infer improved latency, finite-blocklength reliability, or performance under estimated CSI from this idealized fading experiment.

II. SYSTEM MODEL AND EVALUATION OBJECTIVE

A. Two-Round Image Link

For an RGB image $\mathbf{x} \in [0, 1]^{H \times W \times 3}$, a fixed encoder produces $\mathbf{Z} = E_\theta(\mathbf{x}) \in \mathbb{R}^{N \times d}$. Pairing real features forms $d/2$ complex symbols $\mathbf{z}_i^c$ per token. The transmitted token is $\mathbf{s}_i = \mathbf{z}_i^c / \alpha_i$, where $\alpha_i = (2\|\mathbf{z}_i^c\|^2/d + \epsilon)^{1/2}$ and $\epsilon = 10^{-8}$. For $N = 256$, $d = 32$, and $H = W = 256$, each round carries $B = Nd/2 = 4096$ complex payload symbols, giving a payload channel-bandwidth ratio of $1/48$.

Round $r \in \{1, 2\}$ yields $\mathbf{y}_r = h_r \mathbf{s} + \mathbf{n}_r$, with $\mathbf{n}_r \sim \mathcal{CN}(0, 10^{-\gamma/10} \mathbf{I})$ at nominal SNR γ dB. One flat-fading coefficient applies to the image. Equalization uses perfect instantaneous CSI and restores the token scales:

$$\tilde{\mathbf{z}}_{r,i}^c = \alpha_i \frac{h_r^* \mathbf{y}_{r,i}}{|h_r|^2 + \epsilon}. \quad (1)$$

The original experiment uses exact scales. Our signaling audit charges $Q_{16}(\alpha_i)$, an IEEE binary16 value, transmitted once and reused if the same payload is retransmitted; the finite-precision check replaces α_i by $Q_{16}(\alpha_i)$ in the receiver restoration only.

The two rounds follow a correlated Rayleigh law,

$$h_2 = \rho_\Delta h_1 + \sqrt{1 - \rho_\Delta^2} e, \quad \rho_\Delta = J_0(2\pi v f_c \Delta / c), \quad (2)$$

where $h_1, e \sim \mathcal{CN}(0, 1)$ independently, v is speed in m/s, $f_c = 5.9$ GHz, and $\Delta = 1$ ms. Speed changes inter-round correlation, not the marginal first-round fading law. Context $\mathbf{c}$ comprises SNR, speed, CSI age and its Jakes correlation, instantaneous gain, and effective SNR. CSI age is a conditioning variable, not simulated estimation error.

A NACK requests the same normalized payload and produces

$$\mathbf{Z}_M = \frac{|h_1|^2 \tilde{\mathbf{Z}}_1 + |h_2|^2 \tilde{\mathbf{Z}}_2}{|h_1|^2 + |h_2|^2}, \quad (3)$$

with numerical denominator protection in code. The decoder D_θ delivers an accepted first-round state or $D_\theta(\mathbf{Z}_M)$. No third round is allowed. ACK/NACK is reliable; this is image-quality HARQ, not CRC-based error detection. The target $P_0 = 24$ dB need not be attained after the last allowed round.

B. Charged Channel Uses and Quality Constraints

Let $a = 1$ denote a NACK. With metadata and feedback efficiencies η_s, η_f measured in information bits per complex channel use, define

$$C(a) = B(1 + a) + \left\lceil \frac{16N}{\eta_s} \right\rceil + (1 + a) \left\lceil \frac{1}{\eta_f} \right\rceil. \quad (4)$$

The last term charges the initial ACK/NACK and a final ACK when a second round occurs. Metadata is reused rather than retransmitted. These are charged uses under ideal reliable signaling, not a complete air-interface budget: pilots, framing, feedback delay, and control decoding errors are excluded.

Relative to a fixed adaptive-HARQ policy A , we evaluate receiver policy F by the design objective

$$\begin{aligned} \min_F \mathbb{E}[C(a_F)] \\ \text{s.t. } \mathbb{E}[P_A - P_F] \leq 0.02 \text{ dB}, \quad \mathbb{E}[L_F - L_A] \leq 0.003, \end{aligned} \quad (5)$$

where P and L are delivered-image PSNR and LPIPS [8]. This is an evaluation criterion, not a claim that our staged training solves a constrained optimum. Neither average tolerance supplies an image-wise outage guarantee. Writing $p_j = \Pr(a_j = 1)$ and $b_f = \lceil 1/\eta_f \rceil$, the exact accounting identity is

$$\mathbb{E}[C(a_A) - C(a_F)] = (B + b_f)(p_A - p_F). \quad (6)$$

Fixed metadata cancels in the absolute difference but dilutes relative savings through the denominator. Thus $p_A - p_F$, $(p_A - p_F)/p_A$, and $(p_A - p_F)/(1 + p_A)$ represent different quantities: an absolute NACK change, a relative reduction in requests, and a relative payload saving.

III. RELIABILITY-ANCHORED FLOWHARQ

A. Masked Transport with an Identity Constraint

A reliability head predicts $q_i = g_q(\tilde{\mathbf{z}}_{1,i}, \mathbf{c})$, and inference selects $m_i = \mathbb{I}[q_i \geq \tau_r]$. During training only, an oracle mask m_i^* selects the largest $\lceil 0.35N \rceil$ clean-to-received token errors and supervises a balanced binary cross-entropy loss. Neither the clean image nor this oracle mask is available to the deployed receiver.

Following conditional flow matching [9], let $\mathbf{u}_i = \mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}$ and sample $t \sim \mathcal{U}[0, 1]$ on the path $\mathbf{z}_i(t) = \tilde{\mathbf{z}}_{1,i} + tm_i^* \mathbf{u}_i$. The velocity network sees this state, time, receiver context, the received tokens, and a pooled anchor from unselected tokens. To avoid domination by large zero-forcing residuals in deep fades, its regression loss is

$$\begin{aligned} \mathcal{L}_{\text{FM}} = \mathbb{E}_t \frac{\sum_i m_i^* d^{-1} \sum_j \rho_{0.1}((v_{\psi,ij} - u_{ij})/s_i)}{\max(1, \sum_i m_i^*)}, \\ s_i = \max(1, \|\mathbf{u}_i\|/\sqrt{d}), \end{aligned} \quad (7)$$

where $\rho_{0.1}$ is smooth- L_1 and s_i is detached. An image L_1 loss on decoded repair aligns transport with the fixed decoder. That branch uses the predicted hard mask with a straight-through gradient; it does not use the oracle mask from velocity supervision.

At inference, $K = 4$ explicit updates start from $\mathbf{Z}^{(0)} = \tilde{\mathbf{Z}}_1$:

$$\mathbf{Z}^{(k+1)} = \mathbf{Z}^{(k)} + \frac{\mathbf{m}}{K} \odot v_\psi(\mathbf{Z}^{(k)}, (k + \frac{1}{2})/K; \tilde{\mathbf{Z}}_1, \mathbf{m}, \mathbf{c}). \quad (8)$$

Mid-interval time inputs do not make this a second-order mid-point solver. The output projection $\bar{\mathbf{z}}_i = (1 - m_i)\tilde{\mathbf{z}}_{1,i} + m_i \mathbf{z}_i^{(K)}$ preserves every unselected latent token exactly. It does not preserve spatial image regions, since decoding mixes tokens. Likewise, the dense Transformer still processes all tokens; masking constrains content changes rather than delivering sparse-computation savings.

B. Calibrated Quality Decision

A quality head estimates log-MSE from the latent, reliability scores, and context. Its converted PSNR prediction $\hat{P}$ is trained on direct and repaired states using log-MSE regression and a boundary loss

$$\mathcal{L}_q^{\text{boundary}} = \text{BCEWithLogits}((P_0 - \hat{P})/T_q, \mathbb{I}[P < P_0]). \quad (9)$$

The true quality on the right is training supervision only. A separate decision-training stage freezes the codec, reliability head, and velocity field. Scalar biases b_A, b_F are subsequently fitted on calibration images, giving $a_A = \mathbb{I}[\hat{P}(\bar{\mathbf{Z}}_1) + b_A < P_0]$ and $a_F = \mathbb{I}[\hat{P}(\bar{\mathbf{Z}}) + b_F < P_0]$. On ACK, FlowHARQ decodes $\bar{\mathbf{Z}}$; on NACK it decodes (3). The baseline uses the direct state on ACK. Separate calibration is necessary because repair changes the predictor's input distribution; calibration does not, by itself, certify a false-ACK bound.

IV. EXPERIMENTAL RESULTS

A. Protocol and Comparisons

The fixed Swin-based codec has 21.83M parameters, while the reliability, velocity, and quality heads add 0.479M. The velocity field has two Transformer blocks, width 128, and four attention heads. Receiver training uses DIV2K [10] random 256×256 crops for 40 epochs, followed by ten quality-head epochs. AdamW uses batch size 2, learning rate 2×10^{-4} for repair and 10^{-4} for the quality-only stage, weight decay 10^{-4} , and gradient clipping at 5. The boundary temperature is 1 dB, with equal regression and classification weights. Training samples SNR uniformly over 0–15 dB, speed over 0–120 km/h, and CSI age over 0–5 ms. Calibration fixes $K = 4$, $\tau_r = 0.8$, $P_0 = 24$ dB, and the quality biases using the first 20 DIV2K validation images. The remaining 80 images are held out; Kodak24 receives no recalibration. Test images are center-cropped to 256×256 RGB. Tests span SNRs $\{0, 3, 6, 9, 12, 15\}$ dB and speeds $\{30, 60, 90, 120\}$ km/h, with CSI age fixed at 2 ms.

The original confirmation comprises three trained receivers and three channel seeds per receiver. A previously locked extension increases the receiver count to ten; it is reported separately, not relabeled as the original experiment. No additional training or seed sweep is performed for this paper revision. Each policy is evaluated on paired images and physical channel observations. Confidence intervals in Table I use paired per-training-run differences with a Student- t interval, averaging channel draws within each run. Repeated channel draws are not counted as independent training runs.

Controlled comparisons comprise direct one-round decoding, always-two-round MRC, repair without HARQ, adaptive HARQ without repair, and FlowHARQ. These isolate the receiver decision with identical codec and channel resources. They are not independent published architectures. Published author-model references are treated separately below because feedback, training, and channel assumptions differ.

TABLE I
PAIRED FLOWHARQ MINUS ADAPTIVE HARQ. POSITIVE SAVING MEANS FEWER NACKS; POSITIVE ΔLPIPS MEANS WORSE PERCEPTUAL DISTANCE. ORIGINAL AND EXTENDED COHORTS ARE SEPARATED. PSNR IS IN DB.

Dataset	Runs	Saving (pp) [95% CI]	ΔPSNR	ΔLPIPS
DIV2K	3	1.296 [0.023, 2.570]	+0.0206	+0.00132
DIV2K	10	1.319 [1.102, 1.537]	+0.0144	+0.00143
Kodak24	3	2.025 [0.708, 3.343]	+0.0284	+0.00128
Kodak24	10	1.453 [1.081, 1.825]	+0.0253	+0.00131

B. Aggregate Gain and Its Decision Mechanism

The original three-run experiment gives a 1.296-percentage-point NACK reduction on DIV2K and 2.025 points on Kodak24. The ten-run extension retains the effect: DIV2K NACK probability changes from 73.970% to 72.651%, and Kodak24 from 71.701% to 70.249%. Mean delivered PSNR changes by +0.0144 and +0.0253 dB, respectively. LPIPS changes by +0.00143 and +0.00131: both are within the 0.003 tolerance, but neither is a perceptual improvement. Fig. 2 gives the SNR dependence rather than presenting the aggregate as a uniform gain.

To locate the effect, partition paired samples by (a_A, a_F) . The identity $p_A - p_F = \Pr(1, 0) - \Pr(0, 1)$ separates saved from newly requested rounds. On DIV2K, 1.4740% of samples save a round and 0.1545% request an extra one, leaving the net 1.3194-point reduction. Both receivers accept 25.8750% and both request retransmission on 72.4965%. In the last group the delivered images are identical, because both use the same MRC observation.

The quality decomposition is equally informative. DIV2K samples accepted by both policies contribute +0.02719 dB to the overall mean PSNR difference; saved-round samples contribute -0.01332 dB, and extra-round samples +0.00054 dB. Thus the net PSNR gain does not mean that skipping a physical observation improves its reconstruction. It combines a quality benefit on already-accepted images with a small quality sacrifice on saved rounds. Kodak24 has the same structure: 1.5856% saved rounds versus 0.1331% extra rounds. This is an average resource–quality tradeoff, not per-image dominance.

The service target exposes a stronger limitation. Define conditional false ACK as $\Pr(P < P_0 \mid a = 0)$ and final outage as $\Pr(P < P_0)$ after the selected delivery branch. In the ten-run extension, conditional false ACK changes from 13.51% to 14.03% on DIV2K and from 50.88% to 50.57% on Kodak24. FlowHARQ's final outage is 70.89% and 82.92%, respectively. These figures include difficult low-SNR fades and show that 24 dB is not a reliably met service level. Lower average NACK cost within mean-quality tolerances is therefore not evidence of reliable early stopping.

C. Signaling Cost and Finite-Precision Audit

The ten-run NACK reductions correspond to 1.784% and 2.026% fewer second-round requests, but only 0.758% and 0.846% fewer payload symbols. With $\eta_s = \eta_f = 0.5$, 256

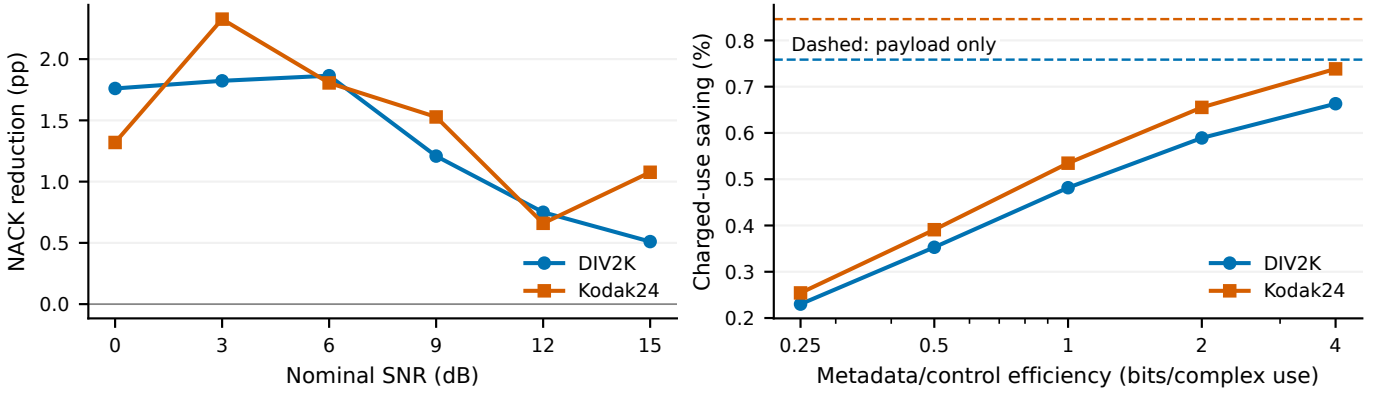


Fig. 2. Frozen ten-run extension, uniformly averaged over images, four speeds, and three channel draws per trained receiver. Left: paired NACK reduction by SNR. Right: relative saving after charging binary16 token scales once and reliable control at $\eta_s = \eta_f$; dashed lines show payload-only savings. The right panel re-accounts archived decisions and is not a new quantized-model performance experiment. All efficiency points are reported.

binary16 scales cost 8192 complex uses once per image, exceeding a single payload round. Equation (4) then gives 0.353% and 0.391% relative charged-use savings. This overhead is a material limitation of the current token-normalized codec. Compressing the scales or changing normalization could improve efficiency, but would change the implemented system and is not credited here.

The original archive records single-scale SSIM, not MS-SSIM. Table II reports a separate frozen-checkpoint audit on Artemis: one existing trained receiver, one channel stream per dataset, all 24 SNR–speed conditions, and paired ideal/binary16 scale restoration. CPU random streams differ from the archived CUDA streams; this is a precision check, not their exact replay.

Quantization changes 0 of 3840 paired adaptive/FlowHARQ decisions on DIV2K and 0 of 1152 on Kodak24. Across all five methods and both datasets, the largest per-image PSNR change is 0.0250 dB. PSNR and MSE alone do not establish perceptual benefit; the measured MS-SSIM and LPIPS must be considered separately. No threshold or weight is refitted using this audit, and it does not replace the multi-run confirmation.

D. One-Step Control and Published Reference Models

An existing parameter-matched, one-step residual control is trained with the same image-alignment objective but with flow time fixed to zero. Its best tested calibration threshold yields -0.0147 dB relative to direct decoding; neither tested threshold passes the predefined nonnegative-PSNR gate. The four-step training recipe yields $+0.0714$ dB in the corresponding earlier calibration study. This supports retaining the measured multistep recipe rather than substituting that one-step control. Different time supervision and threshold selection prevent attributing the difference to integration depth alone, and the result is calibration evidence, not an independent held-out test.

For external context, Table III reports published SwinJSCC [11], NTSCC [12], and CDDM [4] using author weights and a common Kodak24 center-crop AWGN test at 10 dB. PSNR, image MSE, LPIPS, and five-scale MS-SSIM [13]

TABLE II
FROZEN BINARY16 AUDIT UNDER CORRELATED RAYLEIGH FADING. ONE EXISTING RECEIVER AND ONE CHANNEL STREAM PER DATASET, WITH 1920 DIV2K AND 576 KODAK IMAGE–CHANNEL CASES PER METHOD. NO UNCERTAINTY ESTIMATE OR NEW TRAINING IS INFERRED FROM THESE ROWS.

Policy	NACK (%)	PSNR	MSE $\times 10^3$	MS-SSIM	LPIPS
<i>DIV2K</i>					
Direct	0.00	21.319	10.239	0.8000	0.5127
Full HARQ	100.00	21.985	8.587	0.8405	0.4827
Repair only	0.00	21.392	10.165	0.7972	0.5164
Adaptive	74.69	21.912	8.630	0.8379	0.4853
FlowHARQ	73.02	21.948	8.608	0.8371	0.4867
<i>Kodak24</i>					
Direct	0.00	21.470	8.554	0.7797	0.5575
Full HARQ	100.00	22.105	7.142	0.8259	0.5189
Repair only	0.00	21.580	8.409	0.7760	0.5626
Adaptive	74.31	22.033	7.191	0.8220	0.5238
FlowHARQ	71.70	22.070	7.165	0.8209	0.5257

are measured on the same RGB reconstructions. Metadata and final-ACK costs are charged; NTSCC additionally signals token-rate indices. Different rates remain explicit rather than interpolated. These single-round references do not establish FlowHARQ’s superiority under correlated fading. In particular, CDDM’s iterative denoising does not implement our ACK/NACK policy. DeepJSCC-f’s channel-output feedback is also richer than binary HARQ feedback. Our separately archived author-code retraining on DIV2K achieves 26.387 dB, MSE 0.002624, MS-SSIM 0.93746, and LPIPS 0.30644 at two-round forward CBR 1/24 under AWGN 10 dB. This is an adapted-training reference with ideal channel-output feedback, not a matched binary-feedback baseline or a reproduction of its original training dataset.

E. Scope and Limitations

The measured benefit relies on a calibrated receiver and a frozen codec. A different decoder, imperfect CSI, erroneous metadata, or shifted image distribution may invalidate it. The original GPU microbenchmark measures 3.11 ms for decoding and 5.00 ms for repair plus decoding on an RTX

TABLE III

PUBLISHED AUTHOR-MODEL REFERENCES: KODAK24, AWGN 10 DB, TEN CHANNEL DRAWS. CHARGED CBR INCLUDES RELIABLE METADATA/CONTROL; RATES DIFFER. THESE RESULTS ARE NOT MIXED WITH THE RAYLEIGH HARQ COMPARISON.

Model	CBR	PSNR	MSE $\times 10^3$	MS-SSIM	LPIPS
SwinJSCC C32	0.0210	28.39	1.960	0.9518	0.2140
SwinJSCC C64	0.0418	30.82	1.140	0.9732	0.1103
NTSCC Q1	0.0358	28.90	1.630	0.9431	0.2063
NTSCC Q2	0.0515	30.98	0.961	0.9603	0.1279
CDDM C36	0.0236	27.61	2.232	0.9440	0.2169

PRO 6000, but excludes the quality head and communication. It cannot establish lower end-to-end HARQ latency or energy. If additional receiver processing takes δT and a physical round takes T_R , the simplified break-even condition is $\delta T < (p_A - p_F)T_R$; scheduling and overlap require separate measurement. Selective token retransmission and rate-aware flow fine-tuning are outside this paper's validated claim. They should be assessed under a new protocol, not used to enlarge the current contribution without matched evidence.

V. CONCLUSION

FlowHARQ places reliability-anchored latent repair before a quality-based retransmission decision while retaining a common physical MRC fallback. Paired tests support a small aggregate reduction in channel use within stated mean-quality tolerances. Decision decomposition shows where the reduction comes from, and explicit scale signaling substantially reduces its relative size. Receiver refinement is therefore a plausible additional action in HARQ, but its practical value depends on side-information efficiency, calibrated decisions, and the cost of computation relative to a physical round.

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